

A Comprehensive Recap of Foundational Concepts of CS555

Key topics: Statistical foundations, regression, diagnostics, hypothesis testing

Statistical Foundations

Confidence Intervals

- A range of values likely containing the true parameter with a certain level of confidence.
- Example: Call center wait times (30.9 ± 3 minutes, 95% confidence).
- Importance:
 - Provides a range rather than a single value for better decision-making.
 - Acknowledges uncertainty in estimates, showing how much they may vary.

Hypothesis Testing

- Null Hypothesis (H_0): Assumes no effect or difference exists in the population.
- Alternative Hypothesis (H_a): Contradicts H_0 , indicating an effect or difference.
- Examples:
 - One-Sided Test: Testing if a weight loss program reduces weight ($H_0: \mu = 0$, $H_a: \mu < 0$).
 - Two-Sided Test: Testing if water chemical levels differ from normal ($H_0: \mu = 15$ ppm, $H_a: \mu \neq 15$ ppm).

Hypothesis Testing: Z-Test and T-Test

- **Z-Test:**

- Used when population variance is known or $n \geq 30$.
- Example: Call center wait times.

- **T-Test:**

- Applied when population variance is unknown or $n < 30$.

- The population variance (σ^2) is unknown.
- The data is approximately normally distributed (especially important for small samples).
- For two-sample t-tests, the samples should be independent and come from populations with equal variances (if using the pooled t-test).
- **Clarification on $n < 30$:**
- While the t-test is recommended for smaller sample sizes ($n < 30$), it can also be used for larger samples if the population variance is unknown.
- For $n \geq 30$, the central limit theorem ensures that the sample mean distribution approximates normality, so a z-test can be used if the population variance is known.
- Types:
 - One-sample, Two-sample, and Paired t-tests.
 - Example: Comparing test scores of students in two teaching methods

Linear Regression

- **Simple Linear Regression (SLR):**

- Formula: $y = \beta_0 + \beta_1 x$
- y : Dependent variable (outcome).
- x : Independent variable (predictor).
- β_0 : Intercept, the value of y when $x=0$.
- β_1 : Slope, the rate of change in y for each unit increase in x .

- Example: Predicting exam scores based on study hours.

- **Assessing Fit**

- Coefficient of Determination (R^2):

- Indicates how well the independent variable explains the variance in the dependent variable.
- Values range from 0 (no fit) to 1 (perfect fit).

- F-Statistic:

- Tests the overall significance of the regression model.
- Answers the question: Is there a relationship between x and y ?

- **Diagnostics**

- Residual Plots:

- Visualize the difference between observed and predicted values.
- Helps evaluate assumptions (linearity, constant variance, independence).

- Identify outliers and influential points for model refinement.

Advanced Regression Techniques

- **1. Multiple Linear Regression (MLR)**

- Definition: A regression model that incorporates two or more predictors to explain the variability in the dependent variable.
- Key Points:
 - Formula: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k$ where k is the number of predictors.
 - Useful for understanding relationships between multiple independent variables and one dependent variable.
- Diagnosing Multicollinearity:
 - **Variance Inflation Factors (VIFs):**
 - Measures the extent to which predictors are correlated.
 - A high VIF (e.g., >10) indicates severe multicollinearity, which can distort coefficient estimates.

- **ANOVA and Beyond**
- **1. One-Way ANOVA**
- Definition: Tests whether the means of three or more independent groups are significantly different.
- Hypotheses:
 - H_0 : All group means are equal.
 - H_A : At least one group mean is different.
- Example:
 - Comparing weight loss across three diet plans (e.g., Low-carb, Low-fat, Mediterranean).
- **2. Chi-Square Test (optional for CS555)**
- Definition: Tests for independence between two categorical variables.
- Applications:
 - Examining whether customer preferences are independent of gender.
 - Testing if observed data fits a theoretical distribution (goodness-of-fit test).
- **3. Detecting Outliers**
- Residual Analysis:
 - Residuals represent the difference between observed and predicted values.
 - Outliers are points with large residuals that deviate significantly from the model's trend.
- Handling Influential Points:
 - Use diagnostics like Cook's Distance to assess the impact of outliers on regression results.
 - Address by removing, adjusting, or transforming the data.

Logistic Regression

- Definition: A regression model for binary outcomes (e.g., yes/no, 0/1).
- Key Points:
 - Predicts the probability of an event occurring using the logistic function.
 - Output is interpreted as the likelihood of the outcome.
 - Example: Predicting the probability of a coronary event based on cholesterol levels, age, and smoking status.
- **Hypothesis Testing**
- Goal: Assess whether predictors are significantly associated with the outcome.
- Steps:
 - Null Hypothesis (H_0): $\beta_1=0$ or $OR = 1$ (no effect).
 - Decision Rule: Reject H_0 if $|z|>1.96$ (for $\alpha=0.05$).
- **3. Model Fit Evaluation**
- Sensitivity (Recall):
 - Proportion of actual positives correctly identified: $Sensitivity = TP / (TP + FN)$
- Specificity:
 - Proportion of actual negatives correctly identified: $Specificity = TN / (TN + FP)$
- **4. ROC Curve and AUC**
- ROC Curve:
 - Plots Sensitivity (True Positive Rate) vs. 1–Specificity (False Positive Rate).
- AUC (Area Under the Curve):
 - Measures model performance.
 - Ranges from 0.5 (random guessing) to 1.0 (perfect classification).